

TEORÍA DE LOS CIRCUITOS II - R4053 - 2025	
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T.P. N° 6 : Síntesis de cuadripolos	
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Ejercicio #1

Sintetice un cuadripolo RC en vacío que cumpla con:

$$\begin{aligned} -Y_{21} &= \frac{s(s+1)}{s+2} \\ Y_{22} &= \frac{s(s+2.25)}{s+2} \\ \frac{V_2}{V_1} &= \frac{(s+1)}{(s+2.25)} \text{ para } I_2 = 0 \end{aligned}$$

1) Sintetice un cuadripolo RC en vacío que cumpla con:

$-Y_{21} = \frac{s(s+1)}{s+2}$ y $Y_{22} = \frac{s(s+2.25)}{s+2}$ tal que $\frac{V_2}{V_1} = \frac{s+1}{s+2.25}$ con $I_2=0$

1a) Observe estructuras de Y_{21} y Y_{22} y plantee cociente de parámetros

→ De $-Y_{21} = \frac{s(s+1)}{s+2}$, tiene "ceros" en $s=0$ y $s=-1$ y polo en $s=-2$

→ De $Y_{22} = \frac{s(s+2.25)}{s+2}$, tiene "ceros" en $s=0$ y $s=-2.25$ y polo en $s=-2$

Dado que los polos y "ceros" sobre eje s y el origen $\Rightarrow$ sistema disipativo

→ $-Y_{21} = \frac{s(s+1)}{s+2} = \frac{s+1}{s+2.25} \cdot \frac{V_2}{V_1} \Big|_{I_2=0}$

2a) Aplica método gráfico, debe comenzar con $\frac{V_2}{V_1} = 1$ puesto que los ceros de Y_{22} producen polos en transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$. Se pretende, además, eliminar los ceros de transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$, utilizando resistencia pasiva o completo según tipo de ceros, polos y ceros.

The diagrams show the pole-zero distribution for Y_{21} and Y_{22} on the s -plane. For Y_{21} , there are zeros at $s=0$ and $s=-1$, and a pole at $s=-2$. For Y_{22} , there are zeros at $s=0$ and $s=-2.25$, and a pole at $s=-2$. The synthesis process involves canceling the zero at $s=0$ in Y_{22} with the pole at $s=0$ in Y_{21} using a resistor. The remaining pole at $s=-2$ in Y_{21} is canceled with the zero at $s=-2$ in Y_{22} using a parallel combination of a resistor and a capacitor. The final circuit diagram shows a series resistor followed by a parallel combination of a resistor and a capacitor, which is then in series with another parallel combination of a resistor and a capacitor.

32) Puntos residuales por cada polo eliminado, o sea, tenemos sistemas de componentes

$$Z_2 = Z_2 - K_0' \cdot \frac{1}{s} \quad \text{donde } s = 1 \Rightarrow Z_2 = 0, K_0' = \text{Residuo de polo removido por cada polo en origen}$$

$$Z_2|_{s=-1} = 0 = Z_2 - K_0' \cdot \frac{1}{s} \Rightarrow 0 = \frac{s+2}{s(s+2.25)} - K_0' \cdot \frac{1}{s}$$

$$+K_0' \cdot \frac{1}{s} = \frac{s+2}{s(s+2.25)} \Rightarrow K_0' = \frac{s+2}{(s+2.25)} = \frac{-1+2}{-1+2.25} = \frac{1}{1.25} = \frac{4}{5} = K_0'$$

$$\text{Sabemos } K_0' \Rightarrow Z_2 = \frac{s+2}{s(s+2.25)} - \frac{4/5}{s} = \frac{s+2}{s(s+2.25)} - \frac{4/5}{s} = \frac{s+2 - 4/5(s+2.25)}{s(s+2.25)} = \frac{s+2 - 4/5s - 0.9}{s(s+2.25)}$$

$$Z_2 = \frac{\frac{1}{5}s + \frac{1}{5}}{s(s+2.25)} = \frac{4/5 \cdot (s+1)}{s(s+2.25)} = Z_2 \quad \wedge \quad Y_2 = \frac{s(s+2.25)}{4/5 \cdot (s+1)}$$

$$Y_4 = Y_2 - \frac{2K_1 \cdot s}{s+1} = \frac{s(s+2.25)}{(1/5)(s+1)} - \frac{2K_1 \cdot s}{s+1}, \quad 2K_1 = \lim_{s \rightarrow -1} Y_2 \cdot (s+1) =$$

$$2K_1 = \lim_{s \rightarrow -1} \frac{s(s+2.25)}{(1/5)(s+1)} \cdot (s+1) = \frac{-1(-1+2.25)}{(1/5)} = \frac{1.25}{1/5} = 6.25 = 2K_1$$

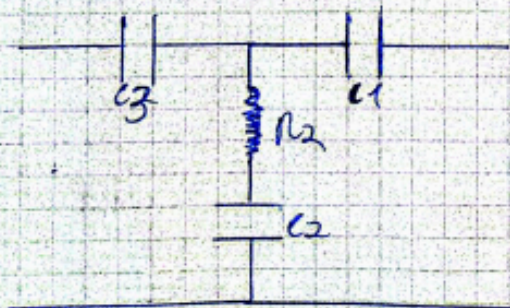
$$\Rightarrow Y_4 = \frac{s(s+2.25)}{(1/5)(s+1)} - \frac{25/4 \cdot s}{s+1} = \frac{s(s+2.25)}{(1/5)(s+1)} - \frac{25/4 \cdot s}{(1/5)(s+1)}$$

$$Y_4 = \frac{s^2 + 2.25s - (5/4)s}{(1/5)(s+1)} = \frac{s^2 + s}{(1/5)(s+1)} = \frac{s(s+1)}{(1/5)(s+1)} = 5s = Y_4 \quad \wedge \quad Z_4 = \frac{1}{5s}$$

$$Z_6 = Z_4 - \frac{K_2}{s}, \quad K_2 = \lim_{s \rightarrow 0} Z_4 \cdot s = \lim_{s \rightarrow 0} \frac{1}{5s} \cdot s = 1/5 = K_2$$

$$Z_6 = \frac{1}{5s} - \frac{1}{5s} = 0$$

42) Planteo circuito con valores de componentes



$$C_1 = \frac{1}{K_0'} = 5/4, \quad C_3 = \frac{1}{K_2} = \frac{1}{1/5} = 5$$

$$C_2 = \frac{2K_1}{0} = \frac{25/4}{1} = 25/4$$

$$R_2 = \frac{1}{2K_1} = 4/25$$

Ejercicio #2

Sintetice un cuádrupolo que cumpla con:

$$Z_{21} = \frac{s}{s^2 + 2}$$

$$Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$$

2) Sintetice un cuádrupolo que cumpla con:

$$Z_{21} = \frac{s}{s^2 + 2} \quad y \quad Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$$

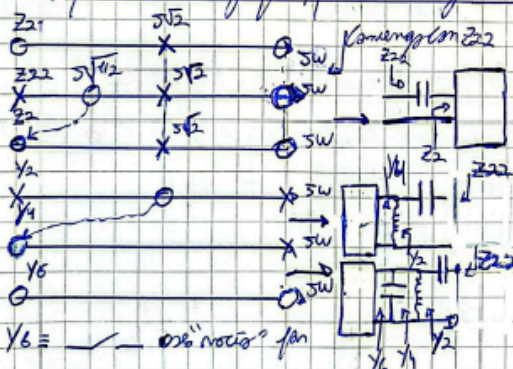
1.9) Observe estructura de Z_{21} y Z_{22}

→ De $Z_{21} = \frac{s}{s^2 + 2}$, tiene "Zero" en $s=0$ y polo en $s = \pm\sqrt{2}$

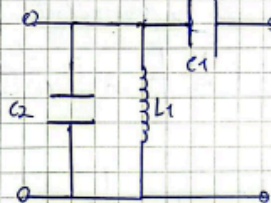
→ De $Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$, tiene "Zero" en $s = \pm\sqrt{2}$ y polo en $s=0$ y $s = \pm\sqrt{2}$

Dado que hay polos y "zeros" sobre eje SW y el origen $\Rightarrow$ sintetizo ma. dispersiva

2.5) Aplico método gráfico, puedo comenzar con Z_{21} o Z_{22} por idénticos circuitos



circuitos resultante:



3.0) Busco residuos por cada polo restante, así obtengo valores de componentes

$$Z_2 = Z_{22} - \frac{K_{01}}{s}, \quad K_{01} = \lim_{s \rightarrow 0} s Z_{22} (s-0) = \lim_{s \rightarrow 0} \frac{1 + 2s^2}{s(s^2 + 2)} \cdot s = \frac{1 + 2 \cdot 0^2}{(0^2 + 2)} = \frac{1}{2} = K_{01}$$

$$Z_2 = \frac{1 + 2s^2}{s(s^2 + 2)} - \frac{1/2}{s} = \frac{1 + 2s^2 - (s^2 + 2)/2}{s(s^2 + 2)} = \frac{1 + 2s^2 - \frac{1}{2}s^2 - 1}{s(s^2 + 2)} = \frac{\frac{3}{2}s^2}{s(s^2 + 2)} = \frac{(3/2)s}{s^2 + 2}$$

$$Z_2 = \frac{(3/2)s}{s^2 + 2}, \quad Y_2 = \frac{s^2 + 2}{(3/2)s}$$

$$Y_4 = Y_2 - \frac{K_{02}}{s}, \quad K_{02} = \lim_{s \rightarrow 0} s Y_2 (s-0) = \lim_{s \rightarrow 0} \frac{s(s^2 + 2)}{(3/2)s} \cdot s = \lim_{s \rightarrow 0} \frac{(2/3)(s^2 + 2)}{(3/2)s} = \frac{2}{3} \cdot \frac{(0^2 + 2)}{(3/2)} = \frac{4}{3}$$

$$Y_4 = \frac{s^2 + 2}{(3/2)s} - \frac{(4/3)}{s}, \quad K_{02} = \frac{4}{3}$$

húsaes

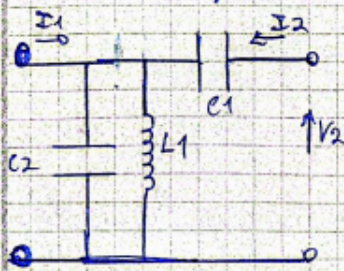
$$Y_4 = \frac{\frac{2}{3}(s^2+2) - 4/3}{s} = \frac{\frac{2}{3}s^2 + \frac{4}{3} - \frac{4}{3}}{s} = \frac{2}{3}s$$

$$Y_5 = Y_4 - K_{00} \cdot s, \quad K_{00} = \lim_{s \rightarrow \infty} Y_4 \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{2}{3}s \cdot \frac{1}{s} = \frac{2}{3} = K_{00}$$

$$Y_5 = \frac{2}{3}s - \frac{2}{3}s = 0$$

4.2) Plantas circuitos con volúmenes de componentes

circuitos con componentes



$$C_1 = \frac{1}{K_{01}} = \frac{1}{1/2} = 2$$

$$C_2 = K_{00} = \frac{2}{3}$$

$$L_1 = \frac{1}{K_{02}} = \frac{1}{4/3} = \frac{3}{4}$$

5.4) Realizo verificación del que obtengo el parámetro Z_{22} trabajando por estos se pueden plantear matrices o usar la definición de parámetro

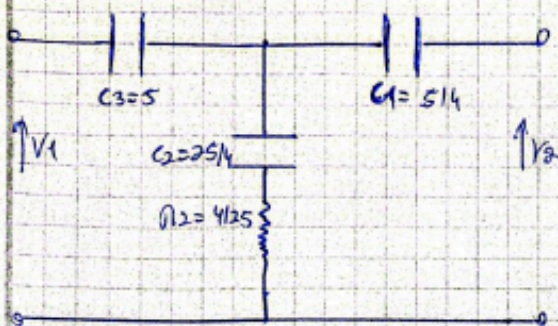
$$Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} = \frac{1}{Y_{L1/C2}} + Z_{E1} = \frac{1}{\frac{1}{\frac{1}{3}s} + K_{00}s} + \frac{K_{01}}{s}$$

$$Z_{22} = \frac{1}{\frac{1}{\frac{3}{4}s} + \frac{2}{3}s} + \frac{1/2}{s} = \frac{1}{\frac{4}{3s} + \frac{2}{3}s} + \frac{1}{2s}$$

$$Z_{22} = \frac{1}{\frac{4 + 2s^2}{3s}} + \frac{1}{2s} = \frac{3s}{4 + 2s^2} + \frac{1}{2s} = \frac{3s \cdot 2s + 2s^2 + 4}{(2s^2 + 4) \cdot 2s}$$

$$Z_{22} = \frac{8s^2 + 4}{(2s^2 + 4)2s} \cdot \frac{1/2}{1/2} \cdot \frac{1/2}{1/2} = \frac{2s^2 + 1}{(s^2 + 2)s} = Z_{22} \text{ Verificado } \checkmark$$

5.9) Verificación



Condición primitiva T

$$V_1 = A \cdot V_2 + B (-I_2)$$

$$I_1 = C \cdot V_2 + D (-I_2)$$

$$\frac{1}{A} = \frac{V_2}{V_1} \Big|_{I_2=0} \quad , \quad \frac{1}{C} = \frac{V_2}{I_1} \Big|_{I_2=0}$$

Plantas producto matrices transpuestas

$$\begin{bmatrix} 1 & Z_{C3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{C2} R_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+Z_{C3} Y_{C2} R_2 & Z_{C3} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

resultado A

$$A = 1 + Z_{C3} Y_{C2} R_2 = 1 + \frac{1}{C_3 \cdot 5} \cdot \frac{1}{\frac{1}{25} + 0.2} = 1 + \frac{1}{\frac{C_3 + 5 C_3 R_2}{C_2}} = 1 + \frac{1}{\frac{5 + 5}{25}} = 1 + \frac{25}{10} = 3.5$$

$$\frac{1}{C_2 R_2} = \frac{1}{25 \cdot 4} = \frac{1}{100} = 0.01, \quad \frac{1}{C_3 R_2} = \frac{1}{5 \cdot 4} = \frac{1}{20} = 0.05$$

$$\Rightarrow 1 + \frac{(5/4)}{5+1} = \frac{5+1+5/4}{5+1} = \frac{5+2.25}{5+1} = A$$

resultado B

$$B = (1 + Z_{C3} Y_{C2} R_2) Z_{C1} + Z_{C3} = A Z_{C1} + Z_{C3} = \frac{5+2.25}{5+1} \cdot \frac{1}{5} + \frac{1}{5} = \frac{5+2.25}{5+1} \cdot \frac{1}{5} + \frac{1}{5} = \frac{5+2.25+5}{5+1} = \frac{12.25}{6} = B$$

resultado Y22

$$Y_{22} = \frac{1}{B} = \frac{(5+1) \cdot 5}{(5+2.25) \cdot 5} = \frac{30}{36.25} = 0.827, \text{ mhos}$$

resultado Y21

$$Y_{21} = -\frac{1}{B} = -\frac{1}{36.25} = -0.0275, \text{ mhos}$$

Ejercicio #3

Sintetice un cuadripolo que cumpla con:

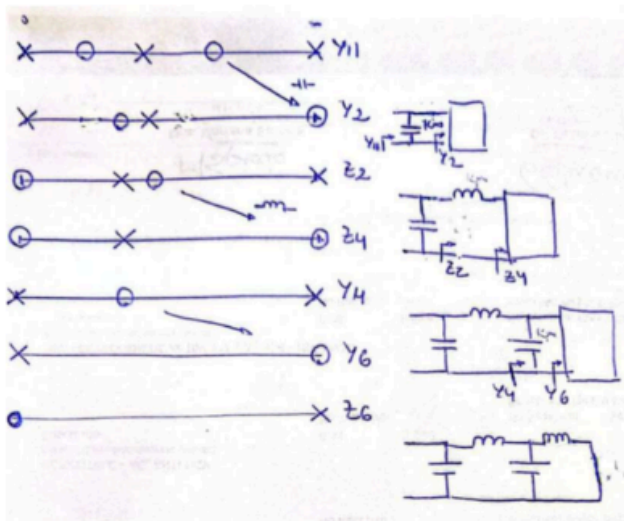
$$Y_{21} = \frac{1}{s(s^2 + 4)}$$

$$Y_{11} = \frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}$$

$$y_{21} = \frac{1}{s(s^2 + 4)}$$

$$y_{11} = \frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}$$

$$\frac{I_2}{I_1} = \frac{y_{21}}{y_{11}} = \frac{\frac{1}{s(s^2 + 4)}}{\frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}}$$



Calculo residuos:

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_{11}}{s} = 1$$

$$y_2 = y_{11} - s = \frac{s^4 + 10s^2 + 9}{s^3 + 4s} - s = \frac{6s^2 + 9}{s^3 + 4s}$$

$$z_2 = \frac{s^3 + 4s}{6s^2 + 9}$$

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{z_2}{s} = \lim_{s \rightarrow \infty} \frac{s^2 + 4}{6s^2 + 9} = \frac{1}{6}$$

$$z_4 = z_2 - \frac{1}{6}s = \frac{s^3 + 4s - \frac{1}{6}s(6s^2 + 9)}{6s^2 + 9} = \frac{4s - \frac{3}{2}s}{6s^2 + 9}$$

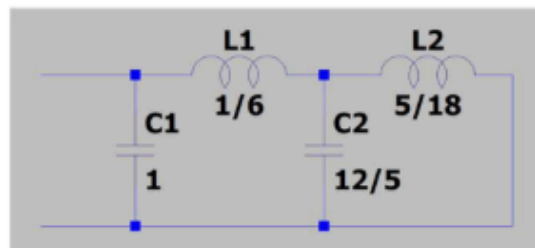
$$z_4 = \frac{5/2 s}{6s^2 + 9}$$

$$y_4 = \frac{6s^2 + 9}{5/2 s}$$

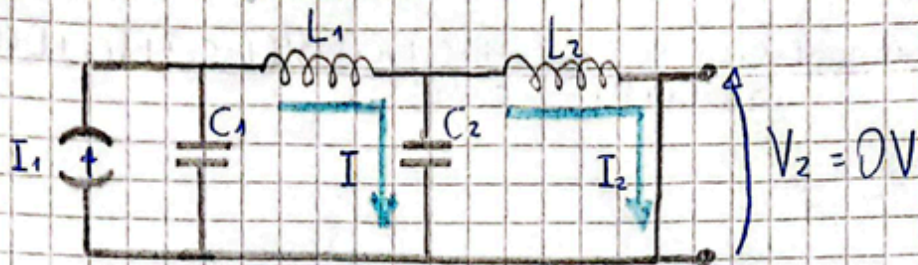
$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_4}{s} = \lim_{s \rightarrow \infty} \frac{6s^2 + 9}{5/2 s^2} = \frac{12}{5}$$

$$y_6 = y_4 - \frac{12}{5}s = \frac{6s^2 + 9 - \frac{12}{5}s \frac{5}{2}s}{\frac{5}{2}s} = \frac{18}{5s}$$

$$z_6 = \frac{5}{18}s$$



Verificación ejercicio 3



$$C_1 = 1 \text{ F} \quad ; \quad L_1 = 1/6 \text{ H}$$

$$C_2 = 12/5 \text{ F} \quad ; \quad L_2 = 5/18 \text{ H}$$

Por mallas:

$$I \cdot \left(\frac{1}{5} + \frac{1}{6} \cdot 5 + \frac{5}{125} \right) - \frac{I_1}{5} - \frac{5}{125} \cdot I_2 = 0 \quad (A)$$

$$I_2 \left(\frac{5}{125} + \frac{5 \cdot 5}{18} \right) - \frac{5}{125} \cdot I = 0 \rightarrow I = I_2 \left(1 + \frac{2 \cdot 5^2}{3} \right) \quad (B)$$

Reemplazo (B) en (A):

$$I_2 \left(1 + \frac{2 \cdot 5^2}{3} \right) \left(\frac{17 + 25^2}{125} \right) - \frac{I_1}{5} - \frac{5}{125} \cdot I_2 = 0$$

$$I_2 \left(\frac{17 + 25^2}{125} + \frac{(34/3)5^2 + (4/3)5^4}{125} - \frac{5}{125} \right) = \frac{I_1}{5}$$

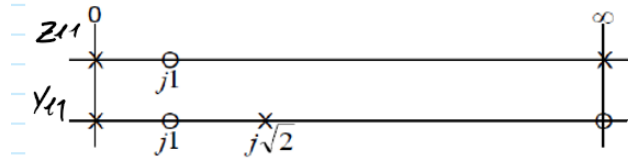
$$I_2 \left(\frac{5^4 + 105^2 + 9}{125} \right) = \frac{3}{4} \cdot \frac{I_1}{5}$$

$$\frac{I_2}{I_1} = \frac{9}{(5^2 + 9)(5^2 + 1)}$$

Ejercicio #4

Sintetice como Lattice simétrico:

4) Sintetice como Lattice simétrico

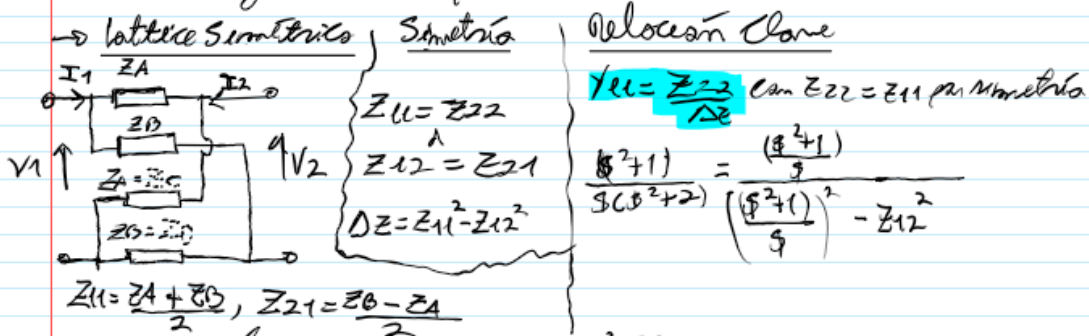


10) Obtenga expresiones de Z_{11} y Y_{11} mirando el diagrama

$$\rightarrow Z_{11} = \frac{(s^2+1)}{s}, \quad Y_{11} = \frac{(s^2+1)}{s(s^2+2)}$$

Recordemos que los polos de transmisión en $s=0$ no se cancelan y que los polos en $s=\infty$ tampoco se cancelan.

2.º) Estrategia de colocación para Lattice simétrico



3) Introduciendo la expresión queda:

$$\frac{(s^2+1)}{s(s^2+2)} = \frac{(s^2+1)}{s \left(\frac{(s^2+1)^2 - Z_{12}^2}{s^2} \right)}$$

$$\frac{(s^2+1)^2 - Z_{12}^2}{s^2} = 1 \Rightarrow (s^2+1)^2 - Z_{12}^2 = s^2+2 \Rightarrow \frac{(s^2+1)^2}{s^2} - (s^2+2) = Z_{12}^2$$

$$Z_{12}^2 = \frac{(s^2+1)^2 - s^2(s^2+2)}{s^2} = \frac{s^4+2s^2+1-s^4-2s^2}{s^2} = \frac{1}{s^2}$$

$$Z_{12} = \frac{1}{s} \Rightarrow Z_{12} = \frac{1}{s} = Z_{21}$$

De $Z_{21} = \frac{Z_B - Z_A}{2} \Rightarrow 2Z_{21} + Z_A = Z_B$ ①

De $Z_{11} = \frac{Z_A + Z_B}{2} \Rightarrow 2Z_{11} - Z_A = Z_B$ ②

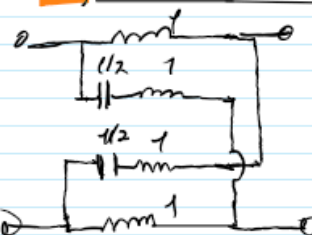
De ① = ② $\Rightarrow 2Z_{21} + Z_A = 2Z_{11} - Z_A$

$$2Z_A = 2Z_{11} - 2Z_{21} \Rightarrow Z_A = Z_{11} - Z_{21}$$

$$Z_A = \frac{(s^2+1)}{s} - \frac{1}{s} = \frac{s^2}{s} = Z_A = \text{Z inductor}$$

Análogamente $Z_B = \frac{1}{s} + s = Z_{\text{capacitor}} + Z_{\text{inductor}}$

4º) circuito propuesto



Ejercicio #5

Conociendo la transferencia:

$$\frac{V_2}{V_1} = \frac{s^2 + 1}{2s^2 + 1}$$

- Obtenga un circuito con 3 inductores y 1 capacitor.
- Obtenga un circuito con 3 capacitores y 1 inductor.

a)

5) a) $\frac{V_2}{V_1} = \frac{s^2 + 1}{2s^2 + 1} = \frac{-Y_{21}}{Y_{22}}$

Plantearmos como polinomio auxiliar de Y_{22} a: $S(S^2 + \frac{3}{4})$

$Y_{22} = \frac{2s^2 + 1}{S(S^2 + \frac{3}{4})}$

The handwritten solution shows the following steps:

- Transfer function: $\frac{V_2}{V_1} = \frac{s^2 + 1}{2s^2 + 1} = \frac{-Y_{21}}{Y_{22}}$
- Plantearmos como polinomio auxiliar de Y_{22} a: $S(S^2 + \frac{3}{4})$
- Expression for Y_{22} : $Y_{22} = \frac{2s^2 + 1}{S(S^2 + \frac{3}{4})}$
- Diagram of a ladder network for Y_{22} with components $j\frac{1}{2}$, $j\sqrt{\frac{3}{4}}$, and $j\omega$.
- Diagram of a ladder network for $1/Y_{22}$ with components $j\frac{1}{2}$, $j\sqrt{\frac{3}{4}}$, and $j\omega$.
- Diagram of a ladder network for Z_2 with components $j\frac{1}{2}$ and $j\omega$.
- Diagram of a ladder network for Y_2 with components $j\frac{1}{2}$ and $j\omega$.
- Diagram of a ladder network for Y_4 with components $j\frac{1}{2}$ and $j\omega$.
- Diagram of a ladder network for Z_4 with components $j\frac{1}{2}$ and $j\omega$.
- Diagram of a ladder network for Z_6 with components $j\frac{1}{2}$ and $j\omega$.
- Calculation of Z_2 : $Z_2 = \frac{1}{Y_{22}} - K_{\infty} \cdot S$
- Calculation of K_{∞} : $K_{\infty} = \frac{S(S^2 + \frac{3}{4})}{S(2S^2 + 1)} \Big|_{S^2 \rightarrow \infty} = \frac{1}{4}$
- Calculation of Z_2 : $Z_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2 + 1} - \frac{1}{4} \cdot S = \frac{S(S^2 + \frac{3}{4}) - \frac{1}{4}S(2S^2 + 1)}{2S^2 + 1}$
- Calculation of Z_2 : $Z_2 = \frac{S^3 + \frac{3}{4}S - \frac{1}{2}S^3 - \frac{1}{4}S}{2S^2 + 1} = \frac{\frac{1}{2}S^3 + \frac{1}{2}S}{2S^2 + 1} \rightarrow Y_2 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)}$

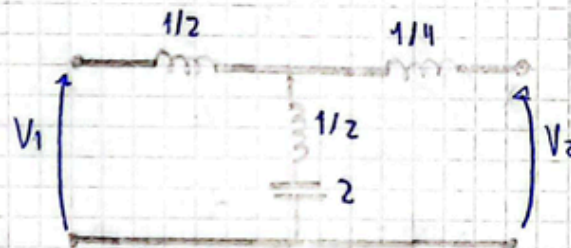
$$Y_4 = Y_2 - \frac{2K_1 S}{S^2 + 1} \quad ; \quad 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1) \cdot (S^2 + 1)}{\frac{1}{2}S(S^2 + 1) \cdot S} = 2$$

$$Y_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{2S^2 + 1 - S^2}{\frac{1}{2}S(S^2 + 1)} = \frac{S^2 + 1}{\frac{1}{2}S(S^2 + 1)} = \frac{1}{\frac{1}{2}S}$$

$$Z_4 = \frac{1}{\frac{1}{2}S}$$

$$Z_6 = Z_4 - K_\infty \cdot S \quad ; \quad K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2}S}{S} = \frac{1}{2}$$

$$Z_6 = \frac{1}{\frac{1}{2}S} - \frac{1}{2} \cdot S = 0$$



Verificación desde los parámetros impedancia:

$$Z_{12} = Z_{21} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

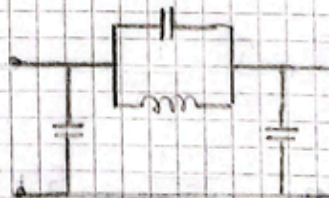
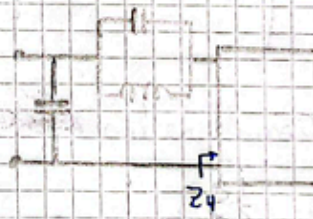
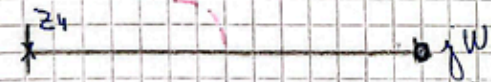
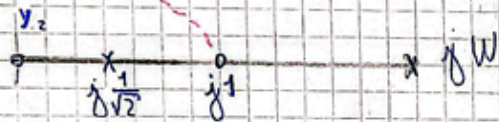
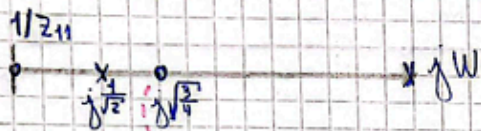
$$Z_{11} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{Z_{21}}{Z_{11}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$

$$b) \frac{V_2}{V_1} = \frac{S^2+1}{2S^2+1} = \frac{Z_{21}}{Z_{11}}$$

Plantamos como polinomio auxiliar de Z_{11} a: $S(S^2 + \frac{3}{4})$

$$Z_{11} = \frac{2S^2+1}{S(S^2+\frac{3}{4})}$$



$$Y_2 = \frac{1}{Z_{11}} - K_{\infty}' \cdot S \quad ; \quad K_{\infty}' = \left. \frac{S(S^2 + \frac{3}{4})}{S(2S^2+1)} \right|_{S^2=-1} = \frac{1}{4}$$

$$Y_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2+1} - \frac{1}{4}S = \frac{\frac{1}{2}S(S^2+1)}{2S^2+1} \rightarrow Z_2 = \frac{2S^2+1}{\frac{1}{2}S(S^2+1)}$$

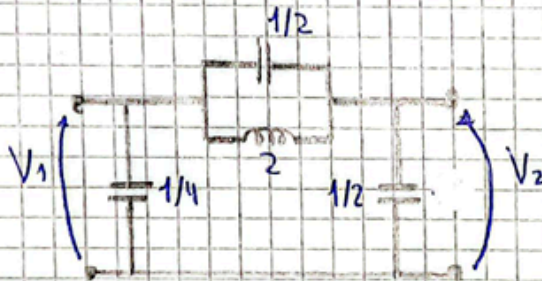
NOTA

$$Z_4 = Z_2 - \frac{2K_1 \cdot S}{S^2 + 1} ; 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1)(S^2 + 1)}{\frac{1}{2}S(S^2 + 1)S} = 2$$

$$Z_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{1}{\frac{1}{2}S} \longrightarrow Y_4 = \frac{1}{2}S$$

$$Y_6 = Y_4 - K_\infty \cdot S ; K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2}S}{S} = \frac{1}{2}$$

$$Y_6 = \frac{1}{2}S - \frac{1}{2}S = 0$$



• Verificación desde los parámetros admitancia:

$$-Y_{21} = -Y_{12} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

$$Y_{22} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{-Y_{21}}{Y_{22}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$

Ejercicio #6

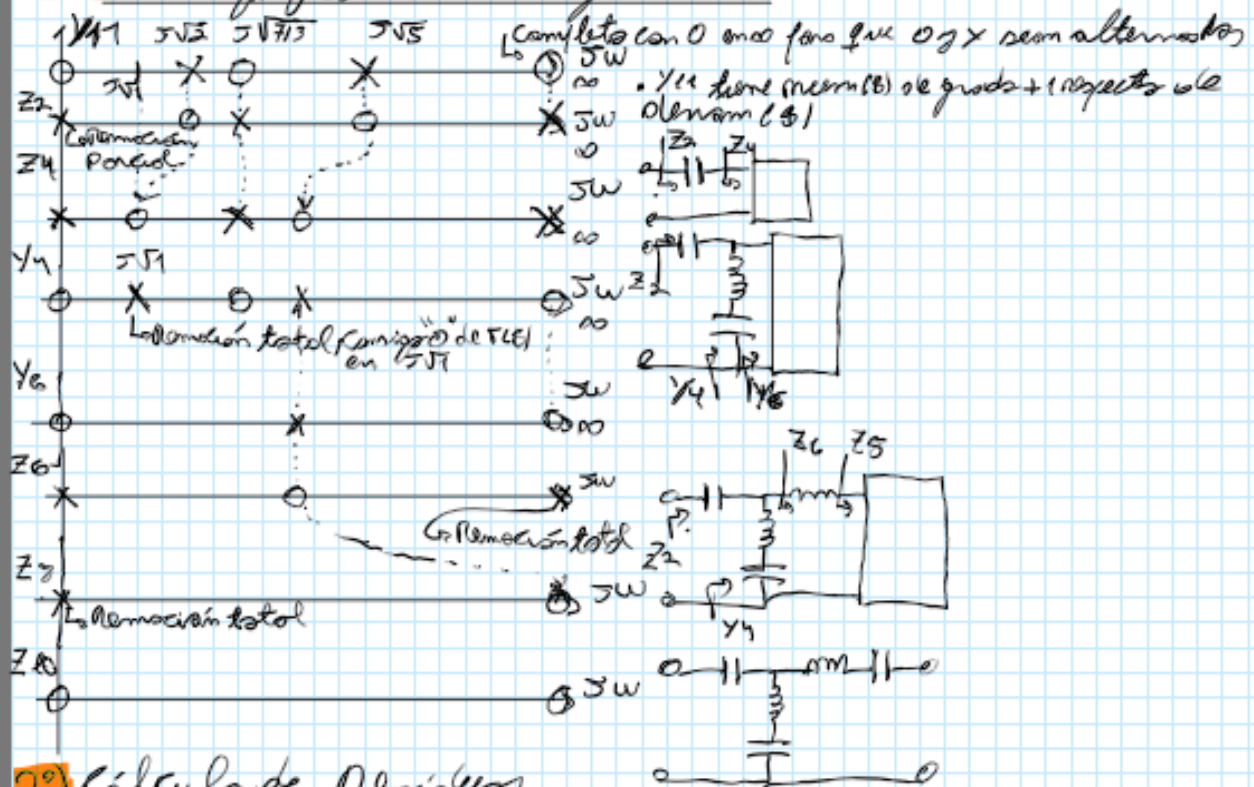
Sintetice un cuadripolo que cumpla con:

$$Y_{11} = \frac{3s(s^2 + 7/3)}{(s^2 + 2)(s^2 + 5)}$$

$$Y_{21} = \frac{s(s^2 + 1)}{(s^2 + 2)(s^2 + 5)}$$

$-\frac{Y_{21}}{Y_{11}} = T(s) = \left. \frac{I_2}{I_1} \right|_{V_2=0} = \frac{s(s^2 + 1)}{3s(s^2 + 7/3)}$, $T(s)$ no tiene polos en $\pm j\omega$ por lo que no se escribe en ecuación de T(s)

1º) Método gráfico de síntesis y remoción



2º) Cálculo de Residuos

• $Z_4(s) = Z_2(s) - \frac{K_0'}{s}$, $Z_4(s) \Big|_{s=0} = 0 = (Z_2(s) - \frac{K_0'}{s}) \Big|_{s=0} \Rightarrow 0 = \frac{(s^2 + 2)(s^2 + 5) - K_0'}{3s(s^2 + 7/3)} \Big|_{s=0}$

$\frac{K_0'}{s} = \frac{(s^2 + 2)(s^2 + 5)}{3s(s^2 + 7/3)} \Rightarrow$ y obteniendo $s^2 = -1 \Rightarrow K_0' = \frac{(-1 + 2)(-1 + 5)}{3(-1 + 7/3)} = 1 = K_0'$

• $Z_4(s) = \frac{(s^2 + 2)(s^2 + 5)}{3s(s^2 + 7/3)} - \frac{1}{s} = \frac{(s^2 + 2)(s^2 + 5) - 3(s^2 + 7/3)}{3s(s^2 + 7/3)} = \frac{s^4 + 7s^2 + 10 - 3s^2 - 7}{3s(s^2 + 7/3)} = \frac{s^4 + 4s^2 + 3}{3s(s^2 + 7/3)}$

$Z_4(s) = \frac{(s^2 + 1)(s^2 + 3)}{3s(s^2 + 7/3)}$, $Y_4(s) = \frac{3s(s^2 + 7/3)}{(s^2 + 1)(s^2 + 3)}$

$$Y_6(s) = Y_4(s) - \frac{2K_1 \cdot s}{(s^2+1)^2}, \quad 2K_1 = \lim_{s \rightarrow -1} Y_4(s) \cdot \frac{(s^2+1)^2}{s} = \lim_{s \rightarrow -1} \frac{3s(s^2+3)}{(s^2+1)(s^2+3)} \cdot \frac{(s^2+1)^2}{s}$$

$$2K_1 = \lim_{s \rightarrow -1} \frac{3(s^2+3)}{(s^2+3)} = \frac{3(-1+3)}{-1+3} = 2 = 2K_1$$

$$Y_6(s) = \frac{3s(s^2+3)}{(s^2+1)(s^2+3)} - \frac{2 \cdot s}{(s^2+1)} = \frac{3s(s^2+3) - 2s(s^2+3)}{(s^2+1)(s^2+3)} = \frac{3s^3 + 9s - 2s^3 - 6s}{(s^2+1)(s^2+3)}$$

$$Y_6(s) = \frac{s(s^2+1)}{(s^2+1)(s^2+3)} = \frac{s}{(s^2+3)} = Y_6(s), \quad \frac{s^2+3}{s} = \frac{36(s)}{s}$$

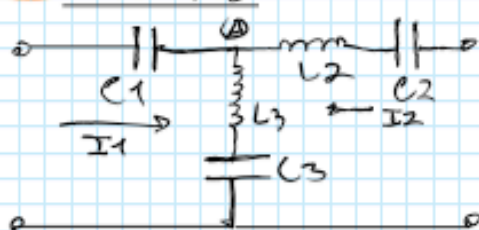
$$Z_7(s) = Z_6(s) - K_{\infty} \cdot s, \quad K_{\infty} = \lim_{s \rightarrow \infty} \frac{Z_6(s)}{s} = \lim_{s \rightarrow \infty} \frac{s^2+3}{s} \cdot \frac{1}{s} = 1 = K_{\infty}$$

$$Z_8(s) = \frac{s^2+3}{s} - 1 \cdot s = \frac{s^2+3-s^2}{s} = \frac{3}{s} = Z_8(s)$$

$$Z_{\infty}(s) = Z_8(s) - \frac{K_0}{s}, \quad K_0 = \lim_{s \rightarrow 0} Z_8(s) \cdot s = \lim_{s \rightarrow 0} \frac{3}{s} \cdot s = 3 = K_0$$

$$Z_{\infty}(s) = \frac{3}{s} - \frac{3}{s} = 0 \Rightarrow \text{cable}$$

32) Circuits



$$Y_{L3,C3} = \frac{1}{\frac{1}{Cs} + sL3} = \frac{1}{\frac{1}{Cs} + sL3} = \frac{sCs}{1 + s^2L3C3}$$

$$Y_{L3,C3} = \frac{sCs}{s^2L3C3 + 1} \cdot \frac{1/(L3C3)}{1/(L3C3)} = \frac{s(1/L3)}{s^2 + \frac{1}{L3C3}}$$

$$\frac{2K_1 \cdot s}{s^2+4} = \begin{cases} \frac{L}{L3} = 2 \\ \frac{1}{L3C3} = 1 \end{cases} \Rightarrow \frac{1}{(1/2)L3} = 1 \Rightarrow 2 = C3$$

$$C1 = \frac{1}{K_0} = 1, \quad C2 = \frac{1}{K_0} = 1/3, \quad L2 = K_0 = 1$$

42) Verifikation

Erinnern dass $\left. -\frac{I_2}{I_1} \right|_{V_2=0} = T(s) = -\frac{Y_{21}}{Y_{11}} \Rightarrow$ Berechnen der Transferfunktion $T(s)$

$$\begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ Y_{L3,C3} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & Z_{L2,C2} \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 + Z_{C1} \cdot Y_{L3,C3} & 1 \cdot 0 + Z_{C1} \\ 0 \cdot 1 + Y_{L3,C3} & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{L2,C2} \\ 0 & 1 \end{bmatrix}$$

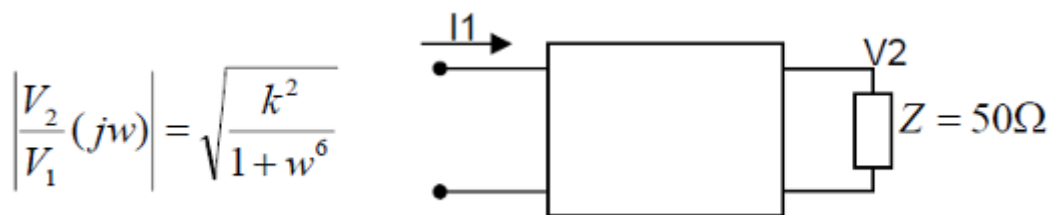
$$[T] \Rightarrow \begin{bmatrix} 1 + Z_{C1} \cdot Y_{L3,C3} + 0 & (1 + Z_{C1} \cdot Y_{L3,C3}) \cdot Z_{L2,C2} + Z_{C1} \\ Y_{L3,C3} \cdot (1 + 0) & Y_{L3,C3} \cdot Z_{L2,C2} + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$D = Y_{L3,C3} \cdot Z_{L2,C2} + 1 = \frac{sCs}{s^2L3C3 + 1} \cdot \frac{1}{(1/2)L3} + 1 = \frac{s \cdot 2}{s^2 + 1} \cdot \left(\frac{3}{s} + 1 \right) + 1 = \frac{2s}{s^2 + 1} \cdot \frac{3 + s^2}{s} + 1 =$$

$$D = \frac{2(3 + s^2)(s^2 + 1)}{s^2 + 1} = \frac{6 + 2s^2 + s^2 + 1}{s^2 + 1} = \frac{3s^2 + 7}{s^2 + 1} = \frac{3 \cdot (s^2 + 1)}{s^2 + 1} = 3 \cdot \frac{(s^2 + 1)}{(s^2 + 1)} = 3 \cdot 1 = 3 \checkmark$$

Ejercicio #7

Sintetice la siguiente transferencia no disipativa:



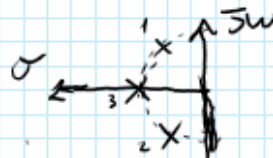
19) Verifiquemos que es la transferencia de un Butter por cojas.

$$\rightarrow \text{Si } B = 5w \Rightarrow \frac{B}{5}, \text{ y } |H(w)|^2 = \frac{1}{1+w^2m}, \quad \varepsilon=1 \text{ con Poles en } \pm j\omega_m \text{ y ceros en } \pm j\omega_c$$

$$\text{Por lo que } \Rightarrow \left| \frac{V_2}{V_1}(s) \right| = \sqrt{\frac{K^2}{1+w^6}} \Rightarrow |H(w)|^2 = \frac{K^2}{1+w^6}$$

$\Rightarrow$ Butter de orden $m=3$.

$$H(s) = \frac{K}{s^3 + 1} \cdot \frac{K}{s+1}$$



$$q_{1,2} = \frac{1}{2 \cos(\varphi_{1,2})}, \quad q_3 = 1, \quad \omega_0 = 1$$

$$\text{Asomados } \frac{V_2}{V_1}(s) = H(s) = \frac{K^2}{s^3 + s^2 + s + 1} = \frac{K^2}{s^3 + 2s^2 + 2s + 1}$$

Por ser Transferencia $\Rightarrow$ Tiene que existir mismo cantidad de polos y ceros $\Rightarrow \arg(\text{denom}(s)) = \arg(\text{num}(s)) \Rightarrow$ Vamos por inspección que hay 3 polos y que deben existir 3 ceros en el s que no se ven en la expresión.

20) Queremos ajustar la expresión $\frac{V_2}{V_1}$ con lo del circuito.

$$\Rightarrow \frac{V_2}{V_1} = \frac{1}{A + B \cdot Y_L + Z_0 \cdot C + Y_L \cdot Z_0 \cdot D} \quad \text{y del diagrama } \Rightarrow \begin{aligned} Z_0 &= 0 \\ Z_L &= 50 \Omega \\ Y_L &= \frac{1}{50} \text{ S} \end{aligned}$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{1}{A + B \cdot Y_L} \quad \text{donde } A = \frac{-Y_{22}}{Y_{21}}, \quad B = \frac{1}{-Y_{21}}$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{1}{\frac{-Y_{22}}{Y_{21}} + \frac{1}{-Y_{21}} \cdot Y_L} = \frac{-Y_{21}}{Y_{22} + Y_L} = \frac{K^2}{s^3 + 2s^2 + 2s + 1}$$

$$\frac{V_2}{V_1} = \frac{1}{\frac{-Y_{22} + 1}{Y_{21}} \cdot \bar{Y}_L} = \frac{-Y_{21}}{Y_{22} + Y_L} = \frac{K^2}{s^3 + 2s^2 + 2s + 1}$$

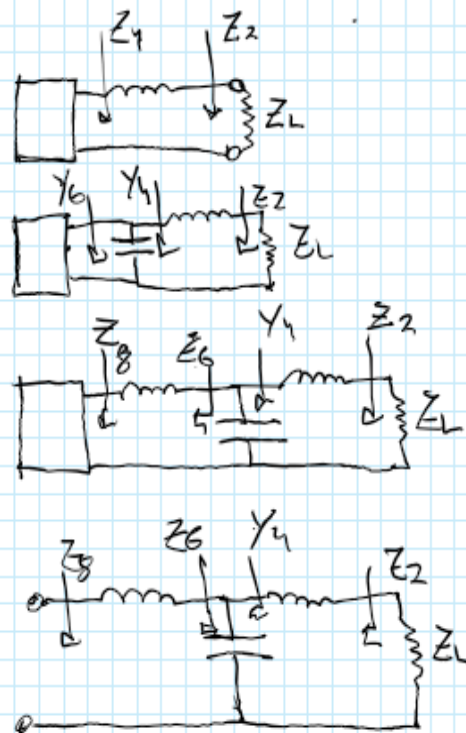
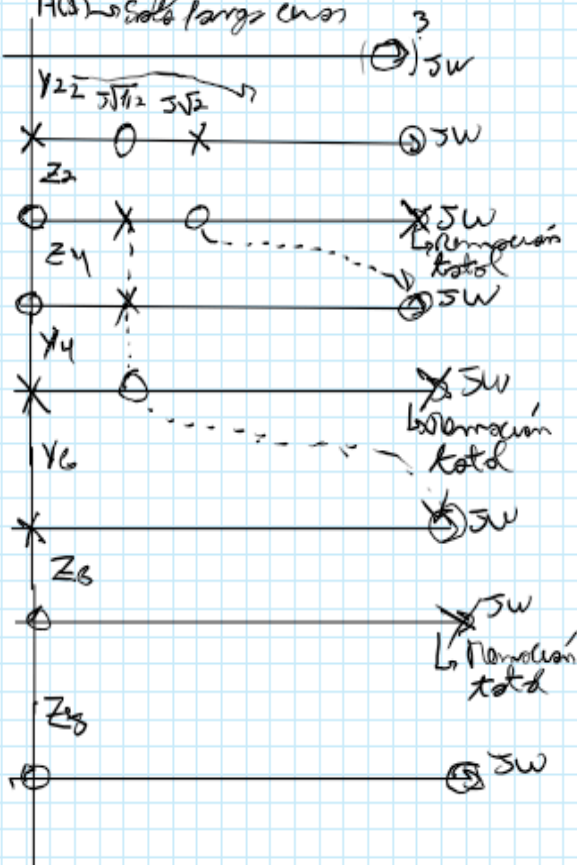
32) Ahora se divide por la parte inferior del denominador de $4(s)$ ya que se demuestra que hay mas soluciones posibles.

$$\Rightarrow \frac{K^2}{s^3 + 2s} = \frac{K^2}{s^3 + 2s} \cdot \frac{s^2 + 1}{s^2 + 1} = \frac{K^2}{s^3 + 2s} \cdot \frac{s^2 + 1}{s^2 + 1}$$

$$\left\{ \begin{array}{l} \frac{K^2}{s^3 + 2s} = -Y_{21} \\ \frac{2s^2 + 1}{s^3 + 2s} = Y_{22} \\ Y_L(\text{manipulado}) = 1 \end{array} \right.$$

42) Sustituimos Y_{22} que es función de excitación, a partir del método gráfico

H(s) solo parte real



6) Cálculo de residuos

$$Y_{22} = \frac{s^3 + 1}{s^3 + 2s}, \quad Z_2 = \frac{1}{Y_{22}} = \frac{s^3 + 2s}{s^3 + 1} \checkmark$$

$$\bullet Z_4(s) = Z_2(s) - K_{0,1} \cdot s, \quad K_{0,1} = \lim_{s \rightarrow \infty} \frac{Z_2(s)}{s} = \lim_{s \rightarrow \infty} \frac{s^3 + 2s}{s^3 + 1} = \frac{1}{2}$$

$$K_{0,1} = \frac{1}{2}$$

$$\bullet Z_4(s) = \frac{s^3 + 2s}{s^3 + 1} - \frac{1}{2} \cdot s = \frac{s^3 + 2s - s^2 - \frac{1}{2}s}{s^3 + 1} = \frac{(3/2)s}{s^3 + 1} = Z_4(s)$$

$$\bullet \frac{2s^2 + 1}{(3/2)s} = Y_4(s)$$

$$\bullet Y_6(s) = Y_4(s) - K_{0,2} \cdot s, \quad K_{0,2} = \lim_{s \rightarrow \infty} \frac{Y_4(s)}{s} = \lim_{s \rightarrow \infty} \frac{2s^2 + 1}{(3/2)s} \cdot \frac{1}{s} = \frac{2}{3/2} = 4/3$$

$$K_{0,2} = 4/3,$$

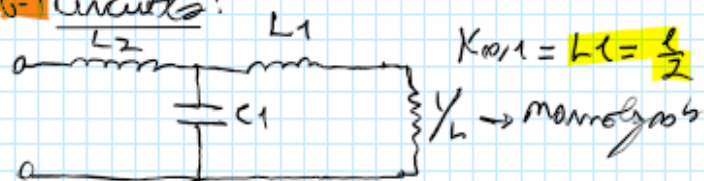
$$\bullet Y_6(s) = \frac{2s^2 + 1}{(3/2)s} - (4/3) \cdot s = \frac{2s^2 + 1 - 2s^2}{(3/2)s} = \frac{1}{(3/2)s} = Y_6(s), \quad (3/2)s = Z_6(s)$$

$$\bullet Z_8(s) = Z_6(s) - K_{0,3} \cdot s, \quad K_{0,3} = \lim_{s \rightarrow \infty} \frac{Z_6(s)}{s} = \frac{(3/2)s}{s} = 3/2$$

$$K_{0,3} = 3/2$$

$$\bullet Z_8(s) = (3/2)s - (3/2)s = 0 \rightarrow \text{corte}$$

6.1) Circuito:



$$K_{0,1} = L_1 = \frac{1}{2}, \quad K_{0,2} = C_1 = 4/3, \quad K_{0,3} = L_2 = 3/2$$

7.0) Verificación

$$\begin{bmatrix} 1 & Z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{c1} & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix} \\ \begin{bmatrix} 1 + Z_2 \cdot Y_{c1} & Z_2 \\ Y_{c1} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix} \\ \begin{bmatrix} 1 + Z_2 \cdot Y_{c1} & (1 + Z_2 \cdot Y_{c1})Z_1 + Z_2 \\ Y_{c1} & Y_{c1} \cdot Z_1 + 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1+Z_{L2}Y_{C1} & (1+Z_{L2}Y_{C1})Z_{L1}+Z_{L2} \\ Y_{C1} & Y_{C1}Z_{L1}+1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix} =$$

$$\begin{bmatrix} (1+Z_{L2}Y_{C1})(1+Z_{L2}Y_{C1})Z_{L1}+Z_{L2} & (1+Z_{L2}Y_{C1})Z_{L1}+Z_{L2} \\ Y_{C1}+(Y_{C1}Z_{L1}+1)Y_L & Y_{C1}Z_{L1}+1 \end{bmatrix}$$

$$\rightarrow A' \Rightarrow \frac{V_2}{V_1} = \frac{1}{A'}$$

$$\frac{V_2}{V_1} = \frac{1}{1+(3/2)\Omega \cdot (4/3)\Omega + ((1+3/2)\Omega(4/3)\Omega)(1/2)\Omega + (3/2)\Omega} \cdot 1$$

$$\frac{V_2}{V_1} = \frac{1}{1+2\Omega^2 + ((1+2\Omega^2) \cdot (1/2)\Omega + (3/2)\Omega)}$$

$$\frac{V_2}{V_1} = \frac{1}{1+2\Omega^2 + (\frac{1}{2}\Omega + \Omega^3 + \frac{3}{2}\Omega)}$$

$$\frac{V_2}{V_1} = \frac{1}{1+2\Omega^2 + 2\Omega + \Omega^3}$$

$$\frac{V_2}{V_1} = \frac{1}{\Omega^3 + 2\Omega^2 + 2\Omega + 1} \quad \text{se demuestra que } K=1 \rightarrow \text{Normalizado}$$

29 Desnormalizando usando $Z_N = 50\Omega = Z_L$ del enunciado

$$L_1 = L_1 N \cdot Z_N = \frac{1}{2} \cdot 50\Omega = 25 \text{ H}$$

$$L_2 = L_2 N \cdot Z_N = \frac{3}{2} \cdot 50\Omega = \frac{150}{2} = 75 \text{ H}$$

$$C_1 = C_1' \cdot Z_N = \frac{4}{3} \cdot 50\Omega = \frac{200}{3} \text{ F}$$

$$R_2 = R_N \cdot Z_N = 1 \cdot 50\Omega = 50\Omega$$

• Se demuestra que si reemplazamos el circuito por el que antes
 $\Rightarrow$ obtenemos $\frac{V_2}{V_1}$ desnormalizado $\Rightarrow \frac{V_2}{V_1} = \frac{\frac{1}{2500}}{\Omega^3 + 2\Omega^2 + \frac{1}{250}\Omega + \frac{1}{2500}}, K^2 = \frac{1}{2500}$

Ejercicio #8

Sintetice el siguiente cuadripolo RC:

$$-\frac{I_2}{I_1} = H \cdot \frac{(s^2 + 5s + 4)}{(s^2 + 8s + 12)}$$

$$Z_{21} = 6H$$



• Comenzamos igualando la expresión $-\frac{I_2}{I_1}$ a la del cuadripolo

$$-\frac{I_2}{I_1} = \frac{1}{A \cdot Y_g \cdot Z_L + C \cdot Z_L + Y_g \cdot D + D}, \text{ donde por simple inspección queda } Y_g = 0, Z_L = 1$$

$$-\frac{I_2}{I_1} = \frac{1}{C \cdot Z_L + D}, C = \frac{1}{Z_{21}}, D = \frac{Z_{22}}{Z_{21}}$$

$$-\frac{I_2}{I_1} = \frac{1}{\frac{1}{Z_{21}} \cdot Z_L + \frac{Z_{22}}{Z_{21}}}, \frac{Z_{21}}{Z_{21}} = \frac{Z_{21}}{Z_L + Z_{22}} = \frac{Z_{21}}{1 + Z_{22}}$$

$$\boxed{-\frac{I_2}{I_1} = \frac{Z_{21}}{1 + Z_{22}}} \Rightarrow \text{Igualando} \Rightarrow \frac{Z_{21}}{1 + Z_{22}} = H \cdot \frac{s^2 + 5s + 4}{(s^2 + 8s + 12)}$$

$$\Rightarrow \frac{6H}{1 + Z_{22}} = H \cdot \frac{(s^2 + 5s + 4)}{s^2 + 8s + 12} \Rightarrow 6 \frac{(s^2 + 8s + 12)}{s^2 + 5s + 4} = (1 + Z_{22})$$

$$\Rightarrow 6 \cdot \frac{(s^2 + 8s + 12)}{s^2 + 5s + 4} = 1 + Z_{22}$$

$$\Rightarrow \frac{6(s^2 + 8s + 12)}{s^2 + 5s + 4} - 1 = Z_{22}$$

$$\Rightarrow \frac{6s^2 + 48s + 72 - s^2 - 5s - 4}{s^2 + 5s + 4} = Z_{22}$$

$$\Rightarrow \boxed{\frac{5s^2 + 43s + 68}{s^2 + 5s + 4} = Z_{22}} \Rightarrow \frac{5(s^2 + 8.6s + 13.6)}{(s+1)(s+4)} = \frac{5 \cdot (s+2.1)(s+6.5)}{(s+1)(s+4)}$$

$$Z_2(s) = \frac{5 \cdot (s+5.6)}{(s+4)}, \quad Z_2(0) = \frac{5(0+5.6)}{0+4} = 7$$

$$Z_4(s) = Z_2(s) - \frac{2K_2}{s+4}, \quad Z_4(s) \Big|_{s=-4} = 0 = \left(Z_2(s) - \frac{2K_2}{(s+4)} \right) \Big|_{s=-4}$$

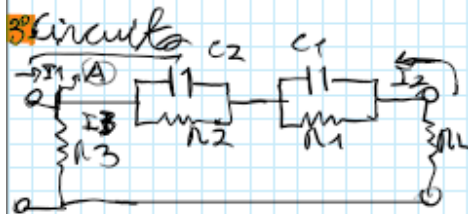
$$\frac{5(s+5.6)}{(s+4)} - \frac{2K_2}{(s+4)} = 0 \Rightarrow \frac{5(s+5.6)}{(s+4)} = \frac{2K_2}{(s+4)} \Rightarrow 5 \cdot (s+5.6) = 2K_2$$

$$\Rightarrow 5(-4+5.6) = 2K_2 \Rightarrow 2K_2 = 8$$

$$Z_4(s) = Z_2(s) - \frac{8}{s+4} = \frac{5(s+5.6)}{(s+4)} - \frac{8}{(s+4)} = \frac{5s+28-8}{s+4} = \frac{5s+20}{s+4} = 5$$

$$Z_4(s) = 5, \quad Y_4(s) = \frac{1}{5}$$

$$Y_4(s) = Y_4(s) - \frac{1}{5} = 0$$



$$Z_{C1R1} = \frac{2K}{s+5}$$

$$Z_{C1R1} = \frac{1}{Y_{C1R1}} = \frac{1}{\frac{1}{s} + \frac{1}{R1}} \cdot \frac{1/C1}{1/C1} = \frac{1/C1}{s + \frac{1}{R1C1}}$$

$$\Rightarrow Z_{C1R1} = \frac{1/C1}{s + \frac{1}{C1R1}} \Rightarrow \begin{cases} 1/C1 = 2K1 = 11 \\ \frac{1}{C1R1} = 5 \end{cases}$$

$$C1 = \frac{1}{55} F$$

$$\frac{1}{C1R1} = 5$$

$$R1 = 11 \Omega$$

$$\Rightarrow Z_{C2R2} = \frac{1/C2}{s + \frac{1}{C2R2}} \Rightarrow \begin{cases} \frac{1}{C2} = 2K2 = 7 \\ C2 = \frac{1}{7} F \\ \frac{1}{C2R2} = 4 \end{cases}$$

$$\Rightarrow R3 = 11 \Omega$$

4) Verificación

Observamos el circuito, aplicamos Kirchhoff ley de voltajes:

$$V_{R3} = V_{R1} + V_{C1R1} + V_{C2R2} \Rightarrow \text{por (A)} \quad I_1 + (-I_2) = I_3, \quad Z' = Z_{C1R1} + Z_{C2R2}$$

$$Z' = \frac{11}{s+5} + \frac{7}{s+4} = \frac{11}{s+5} + \frac{7}{s+4} = \frac{11(s+4) + 7(s+5)}{(s+5)(s+4)} = \frac{18s+51}{s^2+9s+20}$$

$$Z' = \frac{18s+51}{s^2+9s+20} = \frac{18}{s+4} + \frac{51}{s+5}, \quad (I_1 + (-I_2))R3 = I_2 \cdot Z' + I_2 \cdot R1$$

$$I_1 \cdot R3 = I_2 \cdot Z' + I_2 \cdot R1$$

$$(I_1 + I_2)R_3 = I_2 Z' + I_2 R_L \Rightarrow I_1 R_3 = I_2 Z' + I_2 R_L + I_2 R_3$$

$$\Rightarrow \frac{R_3}{Z' + R_L + R_3} = \frac{I_2}{I_1}$$

$$\Rightarrow \frac{5}{18(s + \frac{5}{18}) + 1 + 5} \cdot (s^2 + 5s + 4) = \frac{I_2}{I_1}$$

$$\Rightarrow \frac{5 \cdot (s^2 + 5s + 4)}{18s + 5 + 6(s^2 + 5s + 4)} = \frac{5(s^2 + 5s + 4)}{6s^2 + 48s + 75} = \frac{I_2}{I_1}$$

• Questo verificato y se deduce que $4 = 5/6$

Simulación:

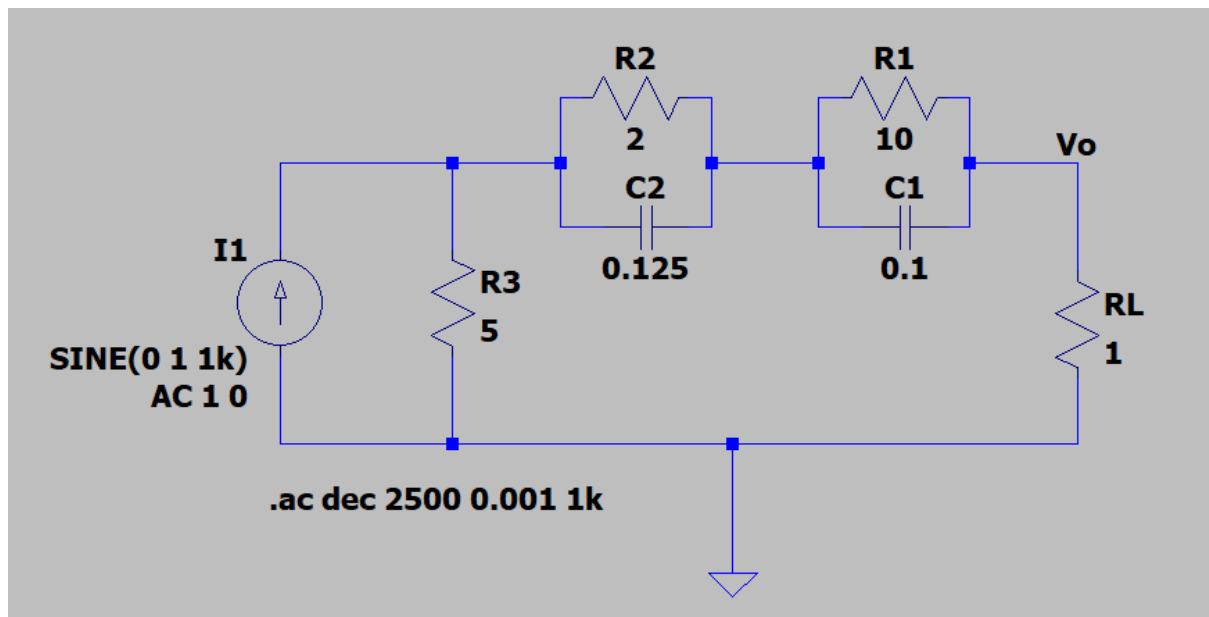
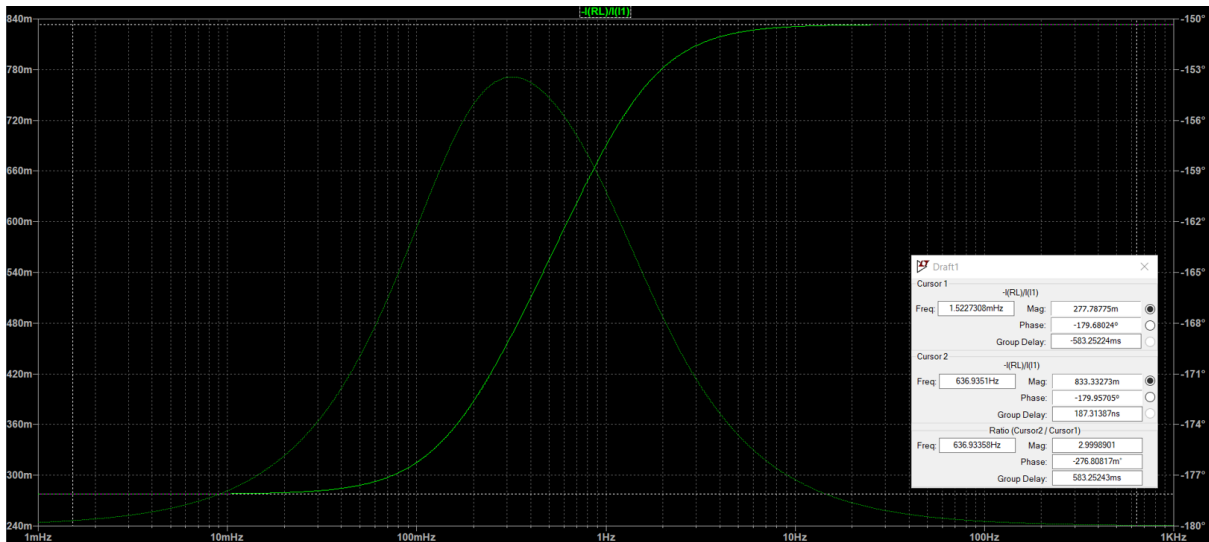


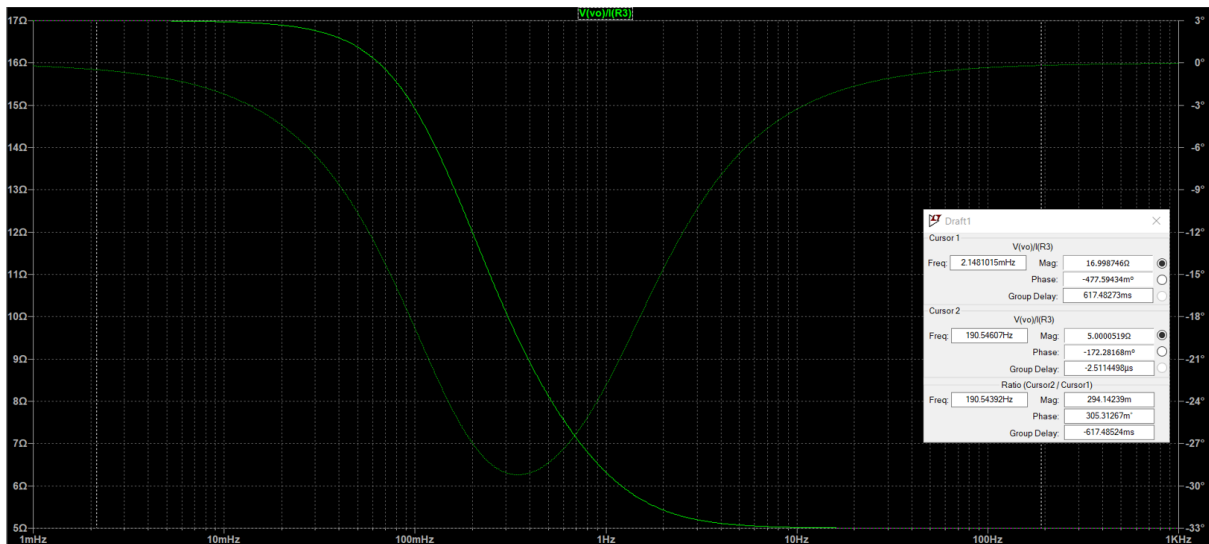
Gráfico 1

Circuito del ejercicio 8.



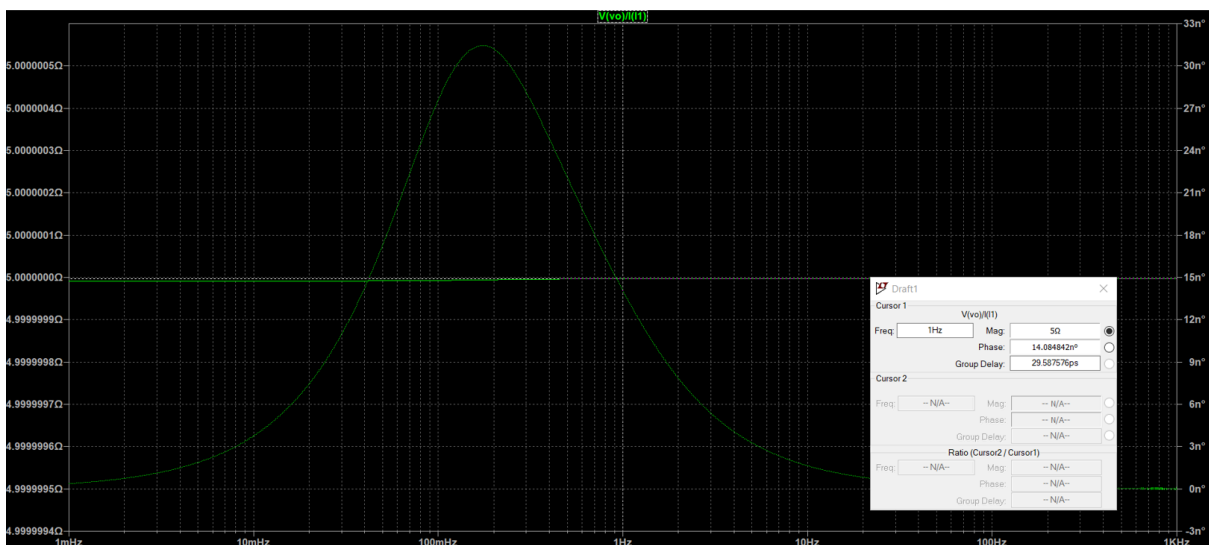
Respuesta en frecuencia de la transferencia de corriente.

Gráfico 2



Respuesta en frecuencia de Z22.

Gráfico 3



Respuesta en frecuencia de Z21.

Gráfico 4

Ejercicio #9

Sintetice mediante un circuito escalera:

$$T(s) = \frac{(s^2 + 4)(s^2 + 9)}{(s^2 + 1)(s^2 + 3)}$$

9) Sintetizar un circuito escalera

$T(s) = \frac{(s^2 + 4)(s^2 + 9)}{(s^2 + 1)(s^2 + 3)}$ $\Rightarrow T = \frac{Z_2}{Y_1} = \frac{Z_2}{Y_1} = -\frac{Y_2}{Z_1}$

Tenemos que usar las funciones de excitación que se representan como sinte-fuer

Necesitamos los polos de T

removemos de forma parcial

al añadir un polo finito, el cero que llega deja un valor reactivo

nuevamente de forma parcial

removemos el polo en adm-tancia

$\Rightarrow Z_2 = Z_{11} - K \frac{s}{s^2 + 4}$ $\wedge Z_2 = Z_{11} / \frac{s^2 + 4}{s^2 + 1} - K \frac{s}{s^2 + 4}$ con $K = \frac{Z_{11}}{s^2 + 4} = \frac{(s^2 + 4)(s^2 + 9)}{s^2 + 1} \cdot \frac{1}{s^2 + 4}$

$Z_2 = \frac{(s^2 + 1)(s^2 + 9)}{s^2 + 2} - \frac{2s}{s^2 + 4} = \frac{s^4 + 9s^2 + 3s^2 + 9}{s^2 + 2} - \frac{2s}{s^2 + 4} = \frac{s^4 + 12s^2 + 9}{s^2 + 2} - \frac{2s}{s^2 + 4}$ $\Rightarrow Z_2 = \frac{s^4 + 12s^2 + 9}{s^2 + 2} - \frac{2s}{s^2 + 4}$

$\Rightarrow Y_2 = \frac{s(s^2 + 2)}{s^2 + 4}$ $\Rightarrow Y_4 = Y_2 - K \frac{s}{s^2 + 4}$ con $K = \lim_{s \rightarrow 0} \frac{s^2 + 4}{s} \cdot \frac{s}{s^2 + 4} = \frac{4}{4} = 1$

$Y_4 = \frac{s(s^2 + 2)}{s^2 + 4} - \frac{s}{s^2 + 4} = \frac{s(s^2 + 1)}{s^2 + 4}$

$= \frac{s(s^2 + 1)}{s^2 + 4} = \frac{s^3 + s}{s^2 + 4} = \frac{s^3}{s^2 + 4} + \frac{s}{s^2 + 4}$

skycolor

$$\Rightarrow Z_4 = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s^3 + \frac{8}{3}s} = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s(s^2 + 4)}$$

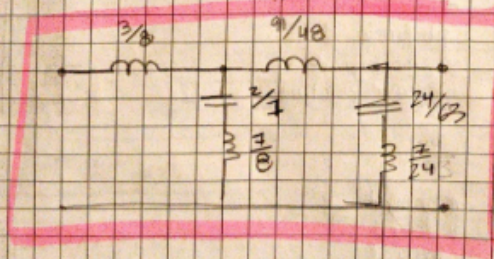
$$\Rightarrow Z_6 = Z_4 - K' \frac{1}{s} \quad \text{on} \quad Z_4 \Big|_{s=-9} - K' \frac{1}{s} \Big|_{s=-9} = 0 \Rightarrow Z_4 \Big|_{s=-9} = K' \frac{1}{s} \Big|_{s=-9}$$

$$\text{on } K' = \frac{Z_4}{\frac{1}{s}} \Big|_{s=-9} = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s(s^2 + 4)} \cdot \frac{1}{s} \Big|_{s=-9} = \frac{\frac{2}{3}(-9 + \frac{5}{3})}{\frac{2}{3}(-9)} = \frac{-\frac{2}{3} \cdot \frac{22}{3}}{-\frac{2}{3} \cdot 9} = \frac{22}{27}$$

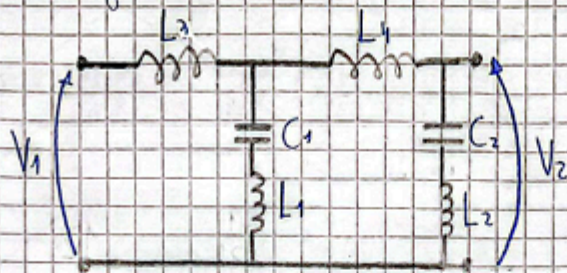
$$\Rightarrow Z_6 = \frac{\frac{2}{3}(s^2 + \frac{5}{3})}{\frac{2}{3}s} - \frac{22}{27} \cdot \frac{1}{s} = \frac{\frac{2}{3}(s^2 + \frac{5}{3}) - \frac{22}{27}s}{\frac{2}{3}s} = \frac{\frac{1}{2}s^2 + \frac{5}{9} - \frac{22}{27}s}{\frac{2}{3}s}$$

$$\Rightarrow Y_6 = \frac{\frac{2}{3}s}{\frac{1}{12}(s^2 + 9)} \rightarrow Y_6 = Y_6 - K \frac{1}{s} \quad \text{on} \quad K = \lim_{s \rightarrow 9} \frac{\frac{2}{3}s}{\frac{1}{12}(s^2 + 9)} \cdot \frac{1}{s} = \frac{24}{7}$$

$$Y_6 = \frac{\frac{2}{3}s}{\frac{1}{12}(s^2 + 9)} - \frac{24}{7} \frac{1}{s} = \frac{\frac{2}{3}s - \frac{24}{7}}{\frac{1}{12}(s^2 + 9)} = 0$$



Verificación ejercicio 9



$$C_1 = 2/7 \text{ F} ; C_2 = 24/63 \text{ F} ; L_3 = 3/8 \text{ H} \\ L_1 = 7/8 \text{ H} ; L_2 = 7/24 \text{ H} ; L_4 = 9/48 \text{ H}$$

$$\begin{pmatrix} 1 + \frac{SL_3}{SL_1 + 1} & SL_3 \\ \frac{1}{SL_1 + 1} & 1 \end{pmatrix} \begin{pmatrix} 1 + \frac{SL_4}{SL_2 + 1} & SL_4 \\ \frac{1}{SL_2 + 1} & 1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = [T]$$

Calculo el parámetro A para obtener luego la transferencia de tensión deseada.

$$A = \left(1 + \frac{SL_3}{SL_1 + 1} \right) \left(1 + \frac{SL_4}{SL_2 + 1} \right) + \frac{SL_3}{SL_2 + 1}$$

$$A = \left(\frac{S^2 L_1 C_1 + S^2 L_3 C_1 + 1}{S^2 L_1 C_1 + 1} \right) \left(\frac{S^2 L_2 C_2 + S^2 L_4 C_2 + 1}{S^2 L_2 C_2 + 1} \right) + \frac{S^2 L_3 C_2}{S^2 L_2 C_2 + 1}$$

$$A = \frac{(S^2 L_1 C_1 + S^2 L_3 C_1 + 1)(S^2 L_2 C_2 + S^2 L_4 C_2 + 1) + (S^2 L_3 C_2) \cdot (S^2 L_2 C_2 + 1)}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

$$A = \frac{S^4 L_1 C_1 L_2 C_2 + S^4 L_1 C_1 L_4 C_2 + S^4 L_3 C_1 L_2 C_2 + S^4 L_3 C_1 L_4 C_2 + S^4 L_3 C_2 L_2 C_2 + S^2 L_2 C_2 + S^2 L_2 C_2 + S^2 L_4 C_2 + S^2 L_4 C_2 + S^2 L_1 C_1 + S^2 L_3 C_1 + 1}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

NOTA

$$A = \frac{S^4(L_1 C_1 L_2 C_2 + L_1 C_1 L_4 C_2 + L_3 C_1 L_2 C_2 + L_3 C_1 L_4 C_2 + L_3 C_2 L_1 C_1) + S^2(L_3 C_1 + L_2 C_2 + L_1 C_1 + L_3 C_1) + 1}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

Reemplazo valores:

$$A = \frac{S^4\left(\frac{7}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{24}{63} \cdot \frac{7}{8} \cdot \frac{2}{7}\right) + S^2\left(\frac{3}{8} \cdot \frac{24}{63} + \frac{7}{24} \cdot \frac{24}{63} + \frac{91}{48} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} + \frac{3}{8} \cdot \frac{2}{7}\right) + 1}{\left(S^2 \frac{7}{8} \cdot \frac{2}{7} + 1\right)\left(S^2 \frac{7}{24} \cdot \frac{24}{63} + 1\right)}$$

$$A = \frac{S^4 \frac{1}{3} + S^2 \frac{4}{3} + 1}{(S^2 \frac{1}{4} + 1)(S^2 \frac{1}{9} + 1)} = \frac{\frac{1}{3}(S^4 + S^2 \cdot 4 + 3)}{\frac{1}{4} \cdot \frac{1}{9}(S^2 + 4)(S^2 + 9)}$$

$$A = 12 \frac{(S^2 + 1)(S^2 + 3)}{(S^2 + 4)(S^2 + 9)}$$

Donde finalmente la transferencia de tensión:

$$A^{-1} = \frac{V_2}{V_1} = \frac{1}{12} \frac{(S^2 + 4)(S^2 + 9)}{(S^2 + 1)(S^2 + 3)}$$

Simulación:

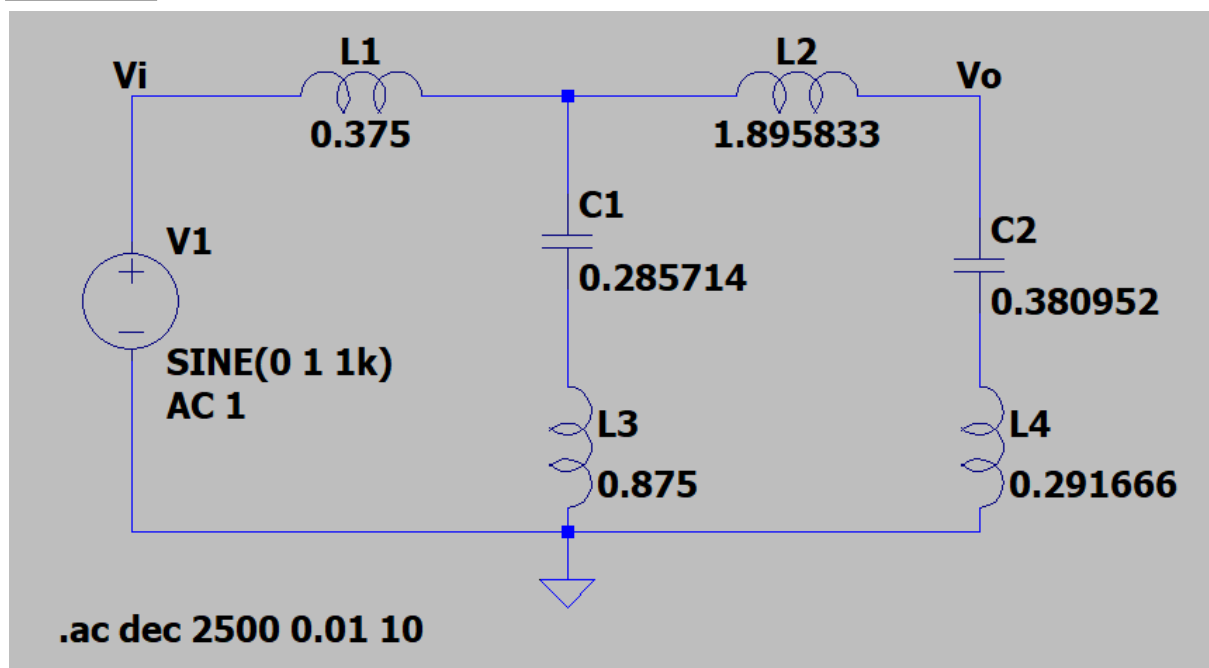


Gráfico 5

Circuito del ejercicio 9.

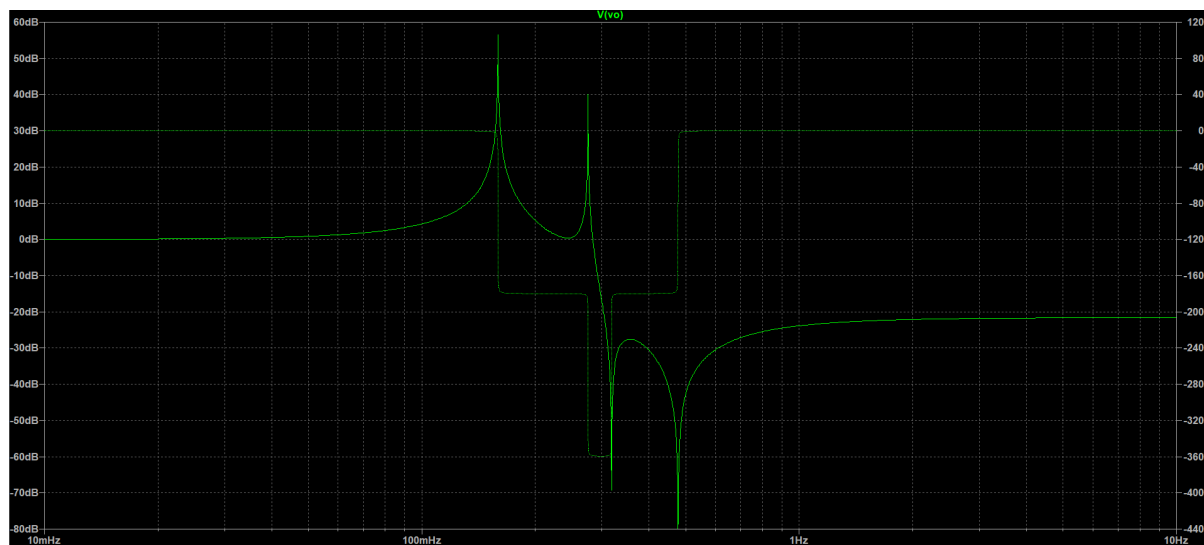


Gráfico 6

Respuesta en frecuencia de la transferencia de tensión.